

EMPLOYMENT

Postdoctoral Research Associate

2023-

Institute for Astronomy, University of Edinburgh

EDUCATION

Ph.D. in Astrophysics

2020–2023

Jodrell Bank Centre for Astrophysics, University of Manchester

Supervisor: Dr. Laura Wolz

Date of thesis defense: Nov. 16th, 2023

Co-Supervisor: Prof. Richard Battye

– Thesis: “Interferometric Neutral Hydrogen Intensity Mapping in the Post-Reionization Universe”

M.Sc. in Astronomy and Astrophysics (Distinction)

2019–2020

Jodrell Bank Centre for Astrophysics, University of Manchester

Supervisor: Dr. Laura Wolz

Co-Supervisor: Prof. Richard Battye

– Thesis: “Interferometric Neutral Hydrogen Intensity Mapping”

B.Sc. in Physics

2015–2019

University of Science and Technology of China (USTC)

Supervisor: Prof. Yi-Fu Cai

STUDENT SUPERVISION

• PhD co-supervision

- Sefa Pamuk, visiting PhD student from IFCA with Dr. José Luis Bernal 2026-
- Gabrielle Autieri, PhD student at SISSA with Dr. Marta Spinelli and Prof. Matteo Viel 2025-
- Jiakang Han, PhD student at UNITO (soon postdoc at KASI) with Prof. Stefano Camera 2024-2026

• Project Supervisor

- Shailesh Devarasu, M.Sc. Theoretical Physics and Mathematical Physics 2026
- Grace Tait, Astrophysics BSc 2026
- Kyle Gillespie, Astrophysics BSc 2025

PUBLICATIONS

† = lead author; * = significant contribution;

Journal articles and preprints under peer-review:

1. J. Townsend et al. (incl. **Z. Chen**), “Measurements of the HI intensity mapping power spectrum at low redshifts with MIGHTEE data: comparison with detected HI galaxies”, submitted to *MNRAS*. arXiv: 2602.23055
2. G. Chen et al. (incl. **Z. Chen**), “Transformer-based Source Detection and Morphological Classification in LOFAR Deep Fields Continuum Images”, submitted to *MNRAS*.
3. MeerKLASS collaboration (incl. **Z. Chen**[†]), “HI intensity mapping with MeerKAT: A measurement of the 21-cm auto-spectrum on cosmological scales”, submitted to *MNRAS*.
4. **Z. Chen**[†], “A quadratic estimator view of the transfer function correction in intensity mapping surveys”, *Mon.Not.Roy.Astron.Soc. Letters* 542 (2025) 1, L1-L6. arXiv: 2504.13070
5. **Z. Chen**[†] et al. (MeerKLASS collaboration), “Emission line stacking of 21cm intensity maps with MeerKLASS: Inference pipeline and application to the L-band deep-field data”, *ApJS* 279 (2025) 19. arXiv: 2504.03908

6. A. Bonaldi et al. (incl. **Z. Chen**), “Square Kilometre Array Science Data Challenge 3a: foreground removal for an EoR experiment”, *Mon.Not.Roy.Astron.Soc.* 543 (2025) 2, 1092-1119. arXiv: 2503.11740
 7. A. Mazumder, L. Wolz, **Z. Chen*** et al. “HI Intensity Mapping with the MIGHTEE Survey: First Results of the HI Power Spectrum”, *Mon.Not.Roy.Astron.Soc.* 541 (2025) 1, 476-493. arXiv: 2501.17564
 8. MeerKLASS collaboration (incl. **Z. Chen***), “MeerKLASS L-band deep-field intensity maps: entering the HI dominated regime”, *Mon.Not.Roy.Astron.Soc.* 537 (2025) 4, 3632-3661. arXiv: 2407.21626
 9. K. Diao, **Z. Chen***, X. Chen and Y. Mao, “Reionization Parameter Inference from 3D Minkowski Functionals of the 21 cm Signals”, *Astrophys.J.* 974 (2024) 141. arXiv: 2406.20058
 10. **Z. Chen**[†], and A. Pourtsidou, “Power spectrum multipoles and clustering wedges during the Epoch of Reionization”, *Mon.Not.Roy.Astron.Soc.* 535 (2024) 3, 2228-2245. arXiv: 2405.05414
 11. **Z. Chen**[†], E. Chapman, L. Wolz and A. Mazumder, “Detecting the HI Power Spectrum in the Post-Reionization Universe with SKA-Low”, *Mon.Not.Roy.Astron.Soc.* 524 (2023) 3, 3724. arXiv: 2302.11504
 12. S. Paul, M. G. Santos, **Z. Chen*** and L. Wolz, “A first detection of neutral hydrogen intensity mapping on Mpc scales at $z \approx 0.32$ and $z \approx 0.44$ ”, *submitted to ApJ letters*. arXiv: 2301.11943
 13. **Z. Chen**[†], L. Wolz and R. Battye, “Towards Optimal Foreground Mitigation Strategies for Interferometric HI Intensity Mapping in the Low-Redshift Universe”, *Mon.Not.Roy.Astron.Soc.* 518 (2023) 2, 2971–2990. arXiv: 2205.07776
 14. **Z. Chen**[†], L. Wolz, M. Spinelli and S. G. Murray, “Extracting HI Astrophysics from Interferometric Intensity Mapping”, *Mon.Not.Roy.Astron.Soc.* 502 (2021) 4, 5259–5276. arXiv: 2010.07985
 15. S. G. Murray, B. Diemer, **Z. Chen*** et al., “TheHaloMod: An online calculator for the halo model”, *Astron.Comput.* 36 (2021) 100487. arXiv: 2009.14066
 16. **Z. Chen**[†], W. Luo, Y.-F. Cai, and E. Saridakis, “New test on general relativity and $f(T)$ torsional gravity from galaxy-galaxy weak lensing surveys”, *Phys. Rev. D* 102 (2020), 104044. arXiv: 1907.12225
 17. B. Li, **Z. Chen***, Y.-F. Cai, and Y. Mao, “Testing the scale-dependent hemispherical asymmetry with the 21-cm power spectrum from the epoch of reionization”, *Mon.Not.Roy.Astron.Soc.* 487 (2019) 4, 5564-5571. arXiv: 1904.04683
 18. **Z. Chen**[†], Y. Xu, Y. Wang and X. Chen, “Stages of Reionization as revealed by the Minkowski Functionals”, *Astrophys.J.* 885 (2019) 23. arXiv: 1812.10333
- Conference proceedings and other articles:
19. A. Mazumder and **Z. Chen**[†] et al., “AASKA II: Interferometric HI intensity mapping with SKA-Mid”, *submitted*.
 20. S. Majumdar et al. (incl. **Z. Chen***), “AASKA II: Cosmology with Intensity Mapping via higher-order statistics”, *submitted*.
 21. M. Spinelli et al. (incl. **Z. Chen**), “AASKA II: Methodological frontiers in 21 cm Intensity Mapping: the treatment of systematics and foreground contamination”, *submitted*.
 22. S. Cunnington and J. Wang (incl. **Z. Chen**), “AASKA II: Single-dish HI intensity mapping with an SKAO precursor: MeerKAT’s Large Area Synoptic Survey (MeerKLASS)”, *submitted*.
 23. L. Wolz et al. (incl. **Z. Chen***), “Advancing Astrophysics with the Square Kilometre Array (AASKA) II: Cosmology with HI Intensity Mapping”, *submitted*.
 24. S. Cunnington et al. (incl. **Z. Chen**), “Revealing cosmological fluctuations in 21cm intensity maps with MeerKLASS: from maps to power spectra”, the Springer Nature Astronomy Prize Awardees Collections (2025), *Astrophys. Space Sci.* 371 (2026) 2, 16. arXiv: 2510.27549
 25. M. Santos, S. Camera, **Z. Chen***, S. Cunnington and J. Fonseca, “Cosmology with ESO–SKAO Synergies”, *the Messenger*, 193 (2024) 20-23. arXiv: 2503.03915

26. **Z. Chen**[†], “Perspective of using multipole power spectrum wedges as summary statistics in future SKA-Low surveys”, contribution to 58th Rencontres de Moriond on Cosmology.

ACADEMIC SERVICE

- Referee for MNRAS, ApJ, PRL, OJA

2022-

COLLABORATION ROLES

MeerKLASS collaboration

I am the co-chair of the PS working group, working on UHF data analysis leading the development in MEER21CM.
I am also a leading member of the interferometric intensity mapping group, working on the analysis pipeline.

SKAO Cosmology Science working group

I am a member of the SKAO cosmology science working group working on HI intensity mapping.

Dark Energy Spectroscopic Instrument

I am a member of the C3 working group working on cross-correlation with 21 cm line.
I am also a member of the GQC Working Group working on mock challenge for the full shape analysis.

SELECTED TALKS

Mining the rainbow: Cosmology in Multicolour via LIM surveys (invited)	07/2025
Title: Towards robust inference of intensity mapping power spectrum	SISSA
A New Era in Astrophysics: preparing for early science with the SKAO	06/2025
Title: Emission line stacking of neutral hydrogen intensity maps: results from MeerKAT	Görlitz, Germany
CASTLE 2024	09/2024
Title: Neutral Hydrogen stacking from MeerKLASS	Tagliolo Monferrato, Italy
Tianlai 2024	07/2024
Title: Clustering wedges of power spectrum multipoles during Epoch of Reionization	HDU, Hangzhou
Moriond 2024	04/2024
Title: Clustering wedges of power spectrum multipoles during Epoch of Reionization	La Thuile, Italy
UK National Astronomy Meeting 2023	07/2023
Title: A first detection of neutral hydrogen intensity mapping	Cardiff University
HITS (HI Intensity Mapping in Trieste) 2022	05/2022
Title: Interferometric Intensity Mapping	SISSA Trieste
SAZERAC 21cm 2022	03/2022
Title: Interferometric Intensity Mapping in the Low-Redshift Universe	Online
UK National Astronomy Meeting 2021	07/2021
Title: Extracting HI Astrophysics from Interferometric Intensity Mapping	University of Bath
2021 SKA Science Conference	03/2021
Title: Extracting HI Astrophysics from Interferometric Intensity Mapping	SKA Organisation
SWIFAR Colloquium (invited)	09/2020
Title: Halo Model, Interferometric Intensity Mapping and HI Shot Noise	Yunnan University